

1. Installing and Verifying curl

Install curl:

```
bash

# Linux (Ubuntu/Debian)
sudo apt update && sudo apt install curl

# Mac
brew install curl

# Verify installation
curl --version
\ \ \

Expected output:
\ \ \

curl 7.81.0 (x86_64-pc-linux-gnu)
Release-Date: 2022-01-05
Protocols: dict file ftp ftps gopher gophers http https imap imaps...
```

2. Basic Usage - Fetch a Webpage

```
bash

curl http://example.com
```

This retrieves and displays the HTML content of the webpage in your terminal.

3. Fetching Data from an API

Get all posts from JSONPlaceholder:

```
bash

curl https://jsonplaceholder.typicode.com/posts
```

Expected output: A JSON array of 100 posts

```
json

[
  {
    "userId": 1,
    "id": 1,
    "title": "sunt aut facere...",
    "body": "quia et suscipit..."
  },
```

```
...  
]
```

Get a single post:

```
bash  
  
curl https://jsonplaceholder.typicode.com/posts/1
```

4. Fetching Only Headers

```
bash  
  
curl -I https://jsonplaceholder.typicode.com/posts  
...
```

Expected output:

```
...  
  
HTTP/2 200  
date: Mon, 09 Feb 2024 10:00:00 GMT  
content-type: application/json; charset=utf-8  
content-length: 27520
```

The `-I` flag fetches only headers (no body), useful for checking server status without downloading content.

5. Making a POST Request

Create a new post:

```
bash  
  
curl -X POST -d "title=foo&body=bar&userId=1" https://jsonplaceholder.typicode.com
```

Expected output:

```
json  
  
{  
  "title": "foo",  
  "body": "bar",  
  "userId": "1",  
  "id": 101  
}
```

With JSON format (recommended):

```
bash  
  
curl -X POST \  
  -H "Content-Type: application/json" \  
  -d '{"title": "foo", "body": "bar", "userId": "1"}'
```

```
-d '{"title": "foo", "body": "bar", "userId": 1}' \  
https://jsonplaceholder.typicode.com/posts
```

6. Common curl Flags

- `-X METHOD` — Specify HTTP method (GET, POST, PUT, DELETE)
- `-H "Header: Value"` — Add custom headers
- `-d "data"` — Send data in request body
- `-I` — Fetch headers only
- `-i` — Include headers in output
- `-v` — Verbose mode (shows full request/response)
- `-o filename` — Save output to file

7. Complete Examples

GET request:

```
bash  
  
curl https://jsonplaceholder.typicode.com/posts/1
```

POST request (create):

```
bash  
  
curl -X POST \  
  -H "Content-Type: application/json" \  
  -d '{"title": "My Post", "body": "Content", "userId": 1}' \  
  https://jsonplaceholder.typicode.com/posts
```

PUT request (update):

```
bash  
  
curl -X PUT \  
  -H "Content-Type: application/json" \  
  -d '{"title": "Updated", "body": "New content", "userId": 1}' \  
  https://jsonplaceholder.typicode.com/posts/1
```

DELETE request:

```
bash  
  
curl -X DELETE https://jsonplaceholder.typicode.com/posts/1
```

8. Formatting Output with jq (Optional)

Install jq:

```
bash
```

```
sudo apt install jq
```

Pretty print JSON:

```
bash
```

```
curl https://jsonplaceholder.typicode.com/posts/1 | jq
```

Extract specific field:

```
bash
```

```
curl https://jsonplaceholder.typicode.com/posts/1 | jq '.title'
```

Summary

- curl is a command-line tool for making HTTP requests
- Basic syntax: `curl [options] URL`
- Use `-X` for HTTP method, `-H` for headers, `-d` for data
- JSONPlaceholder is a free fake API for testing
- The API simulates CRUD operations but doesn't persist data